Data Science Toolbox: Python Programming

Project Report

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Labor Substitution and AI Automation Risk Analysis Using Data Science Techniques

Submitted by

Sachin Raj

Registration No. – 12419763

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Under the Guidance of

Dr. Mrinalini Rana

Discipline of CSE/IT

Lovely School of Computer Science and Technology

Lovely Professional University, Phagwara

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CERTIFICATE

This is to certify that **Sachin Raj**, bearing Registration No. **12419763**, has successfully completed the project titled **"Labor Substitution and AI Automation Risk Analysis Using Data Science Techniques"** in the course **INT 375**, under my guidance and supervision.

To the best of my knowledge, the work presented in this project is the result of his original effort, study, and development.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 22th April 2026

DECLARATION

I, Sachin Raj, student of Data Analysis and Visualization under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 22th April 2026 Sachin Raj

Registration No. 12419763 Signature

Acknowledgement

I would like to express my sincere gratitude to all those who have supported and guided me throughout the completion of this project.

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I am thankful to all the individuals and organizations that provided open-source datasets and resources, which were essential for conducting this analysis.

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Sachin Raj
B.Tech – Computer Science and Engineering
Lovely Professional University, Phagwara

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1. Introduction

1.1 Background

The rapid advancement of Artificial Intelligence (AI) and automated agents has become one of the most significant economic disrupters in the modern world. With the growth of large language models and autonomous agents, the cost of AI inference is decreasing while human labor costs remain static or increase. This dynamic creates a "substitution effect," where industries are increasingly incentivized to replace or augment human labor with AI systems.

Understanding this shift is crucial for businesses, policymakers, and workers. Factors such as hardware capital expenditure, regulatory barriers (moats), and token-generation efficiency all play a role in determining how quickly and effectively a human role can be automated. Therefore, monitoring and analyzing these economic indicators is essential for preparing the workforce of the future.

1.2 Objectives

The primary objective of this project is to perform a comprehensive analysis of the Labor Substitution dataset to extract meaningful insights regarding AI automation risks across various industries. The study is guided by the following specific objectives:

- Understanding the underlying structure and key characteristics of the Labor Substitution dataset
- Executing data cleaning and preprocessing tasks to maintain high standards of data accuracy and consistency
- Conducting exploratory data analysis (EDA) to identify critical trends, patterns, and anomalies in automation risk
- Analyzing the economic relationship between Human Labor Costs and Agent Equivalent Costs
- Generating actionable insights to better understand AI augmentation dynamics and sector-specific vulnerabilities
- Developing a basic predictive model using linear regression to estimate Agent Costs as a function of Inference Costs

1.3 Scope of Study

This study focuses on the analysis of economic data related to AI adoption, cost substitution, and automation risks across various industries. The dataset includes important parameters such as Human Labor Cost, Inference Cost, Automation Risk Index, and Regulatory Moats.

The scope includes exploratory data analysis, visual trend identification, and basic predictive modeling using linear regression. However, the study has certain limitations. It relies on estimations for substitution years (e.g., 2026+) and inference costs, meaning real-world outcomes may vary based on unforeseen technological leaps. Despite this, the analysis provides valuable insights into the vulnerability of specific industries to AI automation.

2. Source of Dataset

2.1 Dataset Description

The dataset utilized in this research project comprises a structured collection of economic indicators focusing on the transition from human labor to AI-driven agents across various professional roles and industrial sectors. It encompasses critical data points regarding the financial expenditures associated with human labor in comparison to the projected operational

costs of AI inference models required to maintain equivalent productivity levels. This quantitative framework facilitates a deeper understanding of the evolving dynamics of automation risk and its varying impact across different economic sectors.

2.2 Features of Dataset

The dataset is comprised of diverse numerical and categorical attributes that facilitate a multidimensional evaluation of labor dynamics:

- **Role_ID & Industry:** These categorical features specify the professional designation and the overarching industrial sector for each observation.
- **Human_Labor_Cost_hr:** Represents the financial expenditure associated with human labor, measured as an hourly rate in US dollars.
- **Tokens_per_Human_Hour:** An estimation of the computational AI tokens necessary to replicate the productivity output of one human labor hour.
- **Inference_Cost_2026:** The projected financial cost per token or inference anticipated for the 2026 fiscal period.
- **Agent_Labor_Equivalent_Cost:** A derived metric representing the total hourly operational cost to deploy an AI agent with equivalent human productivity.
- **Substitution_Elasticity & AI_Augmentation_Factor:** Quantifiable indicators of the feasibility of labor replacement and the degree to which AI enhances residual human performance.
- **Automation_Risk_Index:** A standardized numerical score assessing the susceptibility of a given role to comprehensive AI-driven automation.
- **Regulatory_Moat:** A categorical classification identifying the intensity of legal and compliance barriers that may impede rapid automation adoption.
- **Substitution_Year_Est:** The estimated temporal threshold at which AI implementation becomes economically superior to traditional human labor for the role.

2.3 Data Size and Structure

The dataset is utilized in a structured, tabular format, meticulously organized to facilitate seamless integration with advanced data analysis tools. It encompasses a multidimensional array of numerical attributes, ideal for executing rigorous statistical calculations and predictive regression modeling, alongside categorical classifications that prove essential for the systematic grouping and qualitative interpretation of automation risk severity.

3. EXPLORATORY DATA ANALYSIS (EDA)

3.1 Data Cleaning

Data cleaning is an essential step in the data analysis process, as raw datasets often contain inconsistencies, missing values, and errors that can affect the accuracy of results. Before performing any analysis, it is necessary to ensure that the dataset is clean, consistent, and reliable.

In this project, the dataset was examined carefully to identify any issues such as missing values, duplicate records, and incorrect data formats. These issues were handled systematically to improve the overall quality of the dataset.

One of the primary steps in data cleaning involved checking for missing or null values in different columns. Missing values can lead to incorrect analysis if not handled properly. Depending on the nature of the data, missing values were either removed or appropriately managed to maintain consistency.

Duplicate entries were also checked and removed where necessary to avoid redundancy in the dataset. Duplicate data can distort analysis results and lead to misleading conclusions, especially in large datasets.

Unnecessary or irrelevant artifacts were excluded, and data types were verified to ensure attributes like `Human_Labor_Cost_hr` were treated as integers or floats appropriately.

3.2 Handling Missing Values

Handling missing values is a crucial step in the data preprocessing phase, as incomplete data can significantly affect the accuracy and reliability of the analysis. Missing values may occur due to errors in data collection, equipment malfunction, or unavailability of measurements.

In this project, the dataset was carefully examined to identify missing or null values across all features. The presence of missing values was analyzed to understand their distribution and impact on the dataset.

Different strategies were applied to handle missing values based on the type and importance of the data. For numerical features such as AQI values, missing entries were handled in a way that preserved the overall data distribution. In some cases, rows with missing values were removed to avoid introducing inaccuracies in the analysis.

Incomplete data can significantly affect the accuracy of the analysis. The dataset was examined using the `.isnull().sum()` function in Pandas to identify null values. Missing entries were handled by applying the `.dropna()` method to remove incomplete records, ensuring that the relationships plotted later (like Cost vs. Risk) were based purely on reliable, complete data points.

After applying the necessary steps, the dataset was re-evaluated to ensure that missing values were minimized and the data was suitable for further analysis. This step improved the overall quality and reliability of the dataset, making it more effective for exploratory data analysis and modeling.

3.3 Summary Statistics

Summary statistics provide a concise overview of the dataset by describing its main numerical characteristics. This step is important for understanding the distribution, central tendency, and variability of the data before performing deeper analysis.

Summary statistics were calculated using the `.describe()` function to provide a concise overview of the dataset. Measures such as mean, median, minimum, maximum, and standard deviation were computed. For instance, the mean value of `Human_Labor_Cost_hr` helped establish the baseline economic cost across all industries, while the standard deviation indicated the massive wage disparity between different sectors.

Overall, summary statistics play a crucial role in simplifying large datasets and providing initial insights that guide further exploratory data analysis and modeling.

```
--- Summary Statistics ---
       Human_Labor_Cost_hr  ...  Substitution_Year_Est
count          1200.000000  ...            1200.000000
mean             79.266667  ...            2028.941667
std              38.768390  ...               1.756655
min              20.000000  ...            2026.000000
25%              50.000000  ...            2028.000000
50%              71.000000  ...            2029.000000
75%             103.000000  ...            2030.000000
max             233.000000  ...            2033.000000

[8 rows x 9 columns]

--- Dataset Info ---
<class 'pandas.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 12 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Role_ID                      1200 non-null   str
 1   Industry                     1200 non-null   str
 2   Human_Labor_Cost_hr          1200 non-null   int64
 3   Tokens_per_Human_Hour        1200 non-null   int64
 4   Inference_Cost_2026          1200 non-null   float64
 5   Agent_Labor_Equivalent_Cost  1200 non-null   float64
 6   Substitution_Elasticity      1200 non-null   float64
 7   AI_Augmentation_Factor       1200 non-null   float64
 8   Automation_Risk_Index        1200 non-null   int64
 9   Hardware_CapEx_Sensitivity   1200 non-null   float64
 10  Regulatory_Moat              1200 non-null   str
 11  Substitution_Year_Est        1200 non-null   int64
dtypes: float64(5), int64(4), str(3)
memory usage: 134.5 KB
```

3.4 Outlier Detection

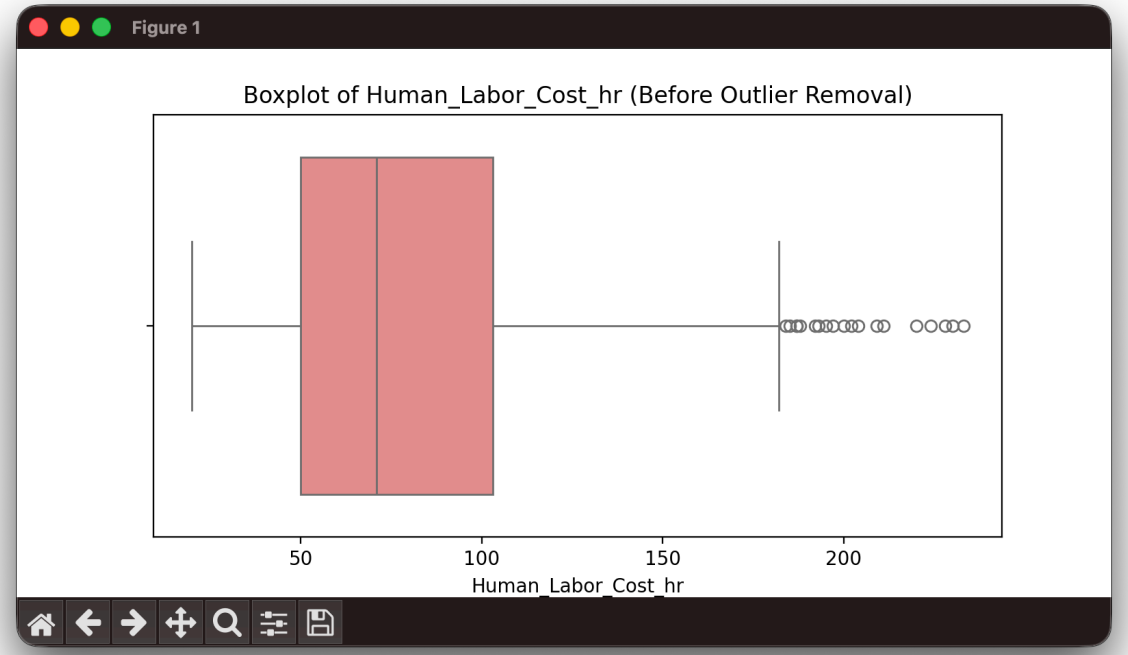
Outliers can significantly influence predictive modeling. In this project, outlier detection was performed on key features like `Human_Labor_Cost_hr`. Using boxplots and the Interquartile Range (IQR) method, data points that deviated considerably from the norm (e.g., hyper-specialized roles with massive hourly rates) were detected and handled to prevent them from skewing the mean and the linear regression line..

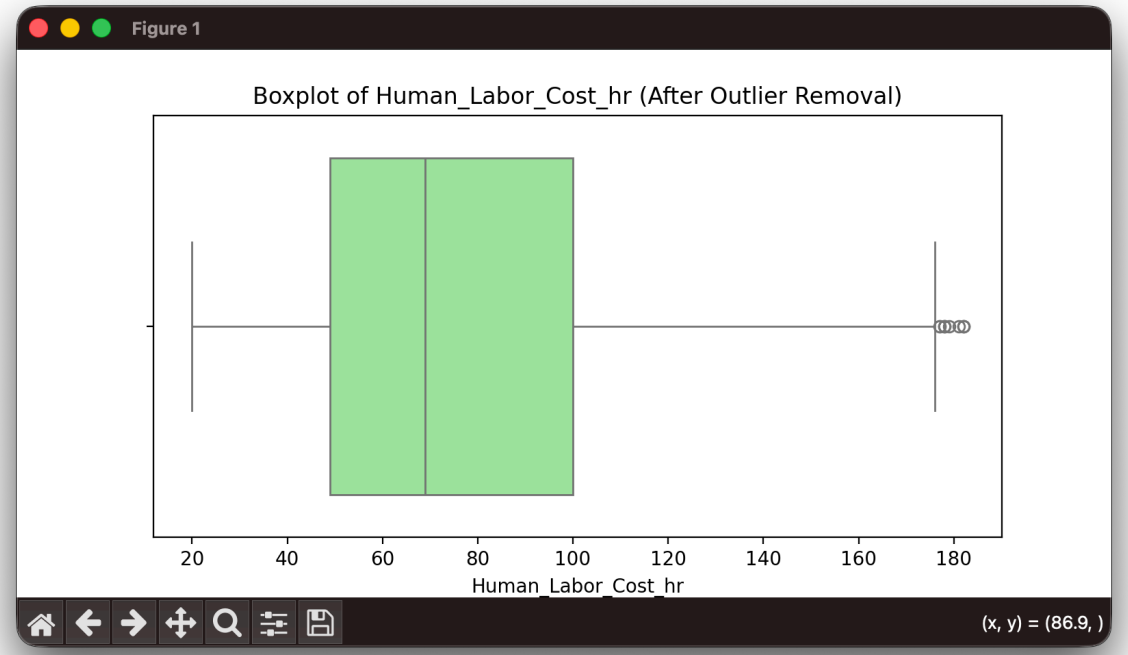
Outliers in AQI values may occur due to extreme pollution events, data recording errors, or rare environmental conditions. Identifying such values is important because they can distort measures like mean and standard deviation, leading to misleading interpretations.

Various techniques can be used for detecting outliers, such as statistical methods and visualization tools. Common approaches include analyzing the range of values, using interquartile range (IQR), and examining graphical representations like box plots.

In this analysis, outliers were carefully examined to determine whether they represented genuine extreme conditions or errors in the dataset. Instead of blindly removing all outliers, a balanced approach was taken to retain meaningful extreme values while eliminating only those that could negatively affect the analysis.

Handling outliers appropriately ensures that the dataset remains both accurate and representative of real-world conditions. This step improves the reliability of subsequent analysis, including statistical evaluation and predictive modeling.





4. ANALYSIS ON DATASET

4.1.1 Introduction

Statistical analysis plays a crucial role in understanding the underlying patterns and relationships within a dataset. It involves the application of mathematical and statistical techniques to summarize, interpret, and draw meaningful conclusions from data.

In this project, statistical analysis is performed on the Labor Substitution dataset to examine the distribution and variation of automation risk across different industries and regulatory environments. By applying statistical methods, it becomes possible to identify trends, compare values, and evaluate the consistency of the data.

The primary focus of this analysis is on key numerical variables such as Human Labor Cost, Automation Risk Index, and Agent Equivalent Cost. These variables provide valuable insights into the level of financial threat posed by AI. Through statistical analysis, it is possible to identify patterns such as whether automation risk is consistently high in certain industries or whether it fluctuates based on regulatory barriers.

This section lays the foundation for further detailed analysis by providing a structured approach to interpreting the data using Python libraries like Seaborn and Matplotlib.

4.1.2 Methods and Formulas

In order to perform statistical analysis on the dataset, several fundamental statistical measures and methods were used. These methods help in summarizing the data and understanding its distribution and variability.

One of the primary measures used is the **mean (average)**, which represents the central value of the dataset. It is calculated by summing all the observations and dividing by the total number of observations.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Another important measure is the **median**, which represents the middle value when the data is arranged in ascending order. The median is less affected by extreme values.

The **standard deviation** is used to measure the spread or variability of the data. It indicates how much the values deviate from the mean.

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

The **minimum and maximum values** are also considered to understand the range of the dataset. Basic statistical comparisons were performed to analyze how Automation Risk values vary across different categorical variables like Industry and Regulatory Moat. These methods provide a strong foundation for analyzing the dataset and support the interpretation of visual results.

4.1.3 Results and Interpretation

The statistical analysis of the dataset revealed significant variation in Human Labor Costs and Automation Risk Index values across different industries. The wide range between minimum and maximum values, along with high standard deviation in costs, indicates that the threat of AI automation is highly inconsistent and varies considerably across sectors.

Geographical and sectoral analysis highlights that certain industries consistently show higher Automation Risk levels, while others maintain lower risk due to specific human-centric requirements. This variation is heavily influenced by factors such as the routine nature of the tasks and the tokens required per human hour.

The distribution of Risk across Regulatory Moat categories (Low, Med, High) shows clear separation, with High regulatory moats having significantly lower automation risks. Overall, the results indicate that AI labor substitution is influenced by multiple factors and varies significantly, emphasizing the need for targeted corporate strategies.

4.1.4 Visualization

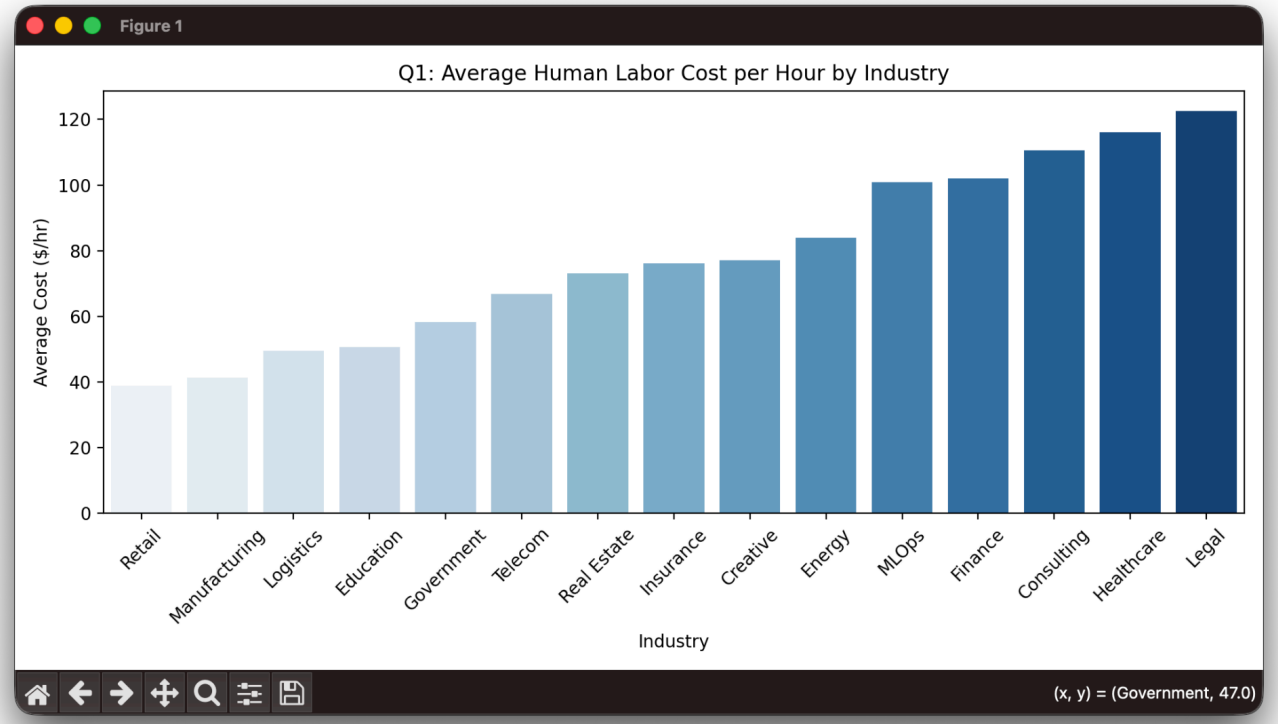
Visualization plays a key role in statistical analysis by providing a clear and intuitive understanding of the data. Graphical representations make it easier to identify patterns, trends, and variations that may not be immediately visible through numerical analysis alone.

In this project, various types of visualizations were used to represent the distribution and variation of variables.

- **Bar charts** were used to compare Average Human Labor Cost and Average AI Augmentation Factor across different industries. This allowed for easy identification of areas with higher or lower financial vulnerabilities.

- **Histograms** were used to analyze the distribution of the Estimated Substitution Year, helping to identify whether the timeline for automation is skewed towards the near future.
- **Scatter plots** were utilized to map relationships between continuous variables, such as Human Labor Cost versus Agent Equivalent Cost.
- **Box plots** were particularly useful in detecting variations across categories, such as mapping Automation Risk Index grouped by Regulatory Moat levels.

Overall, visualizations enhanced the statistical analysis by presenting complex economic data in a simplified and understandable form. They supported the findings obtained from numerical analysis.



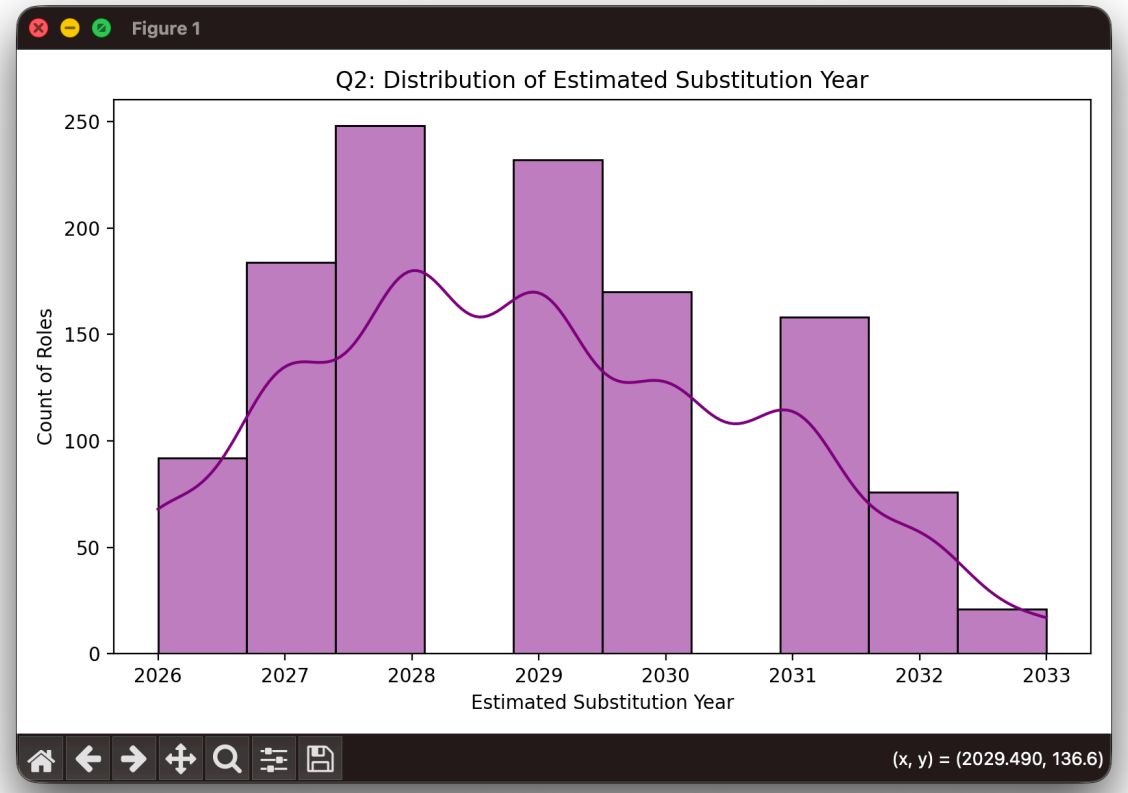
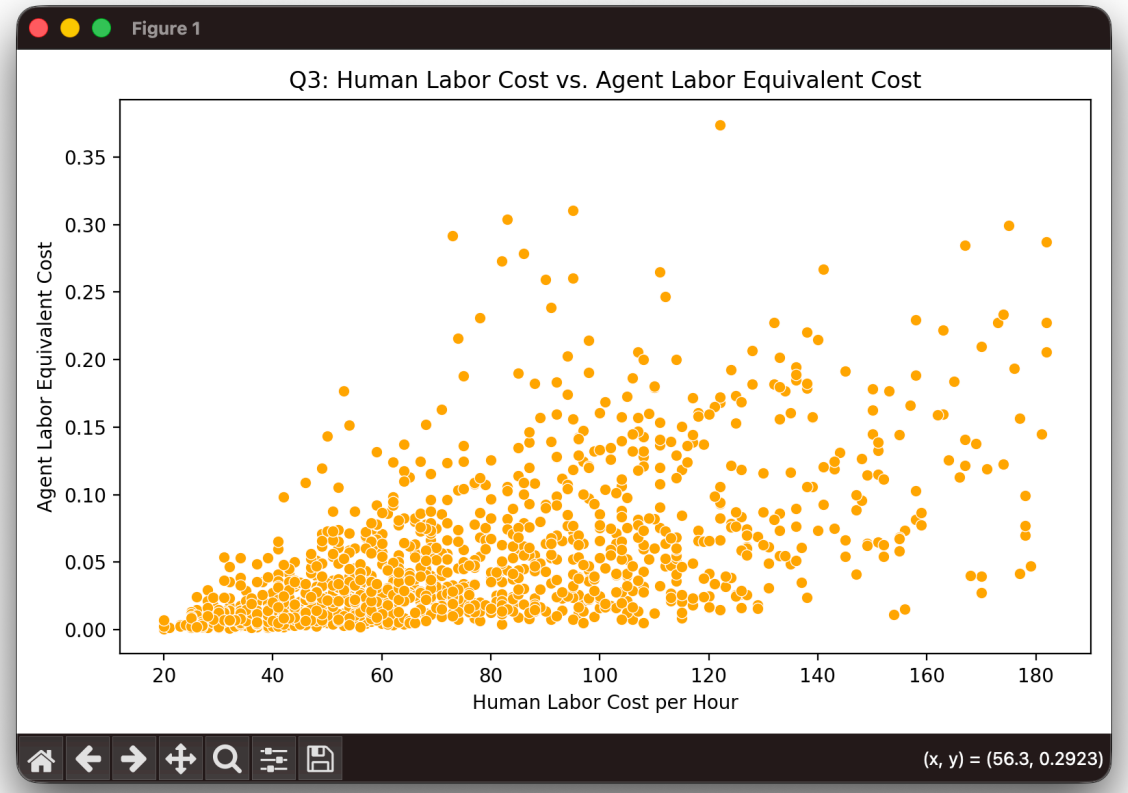


Figure 1
Q2: Distribution of Estimated Substitution Year
Count of Roles
0
50
100
150
200
250
2026
2027
2028
2029
2030
2031
2032
2033
Estimated Substitution Year
(x, y) = (2029.490, 136.6)



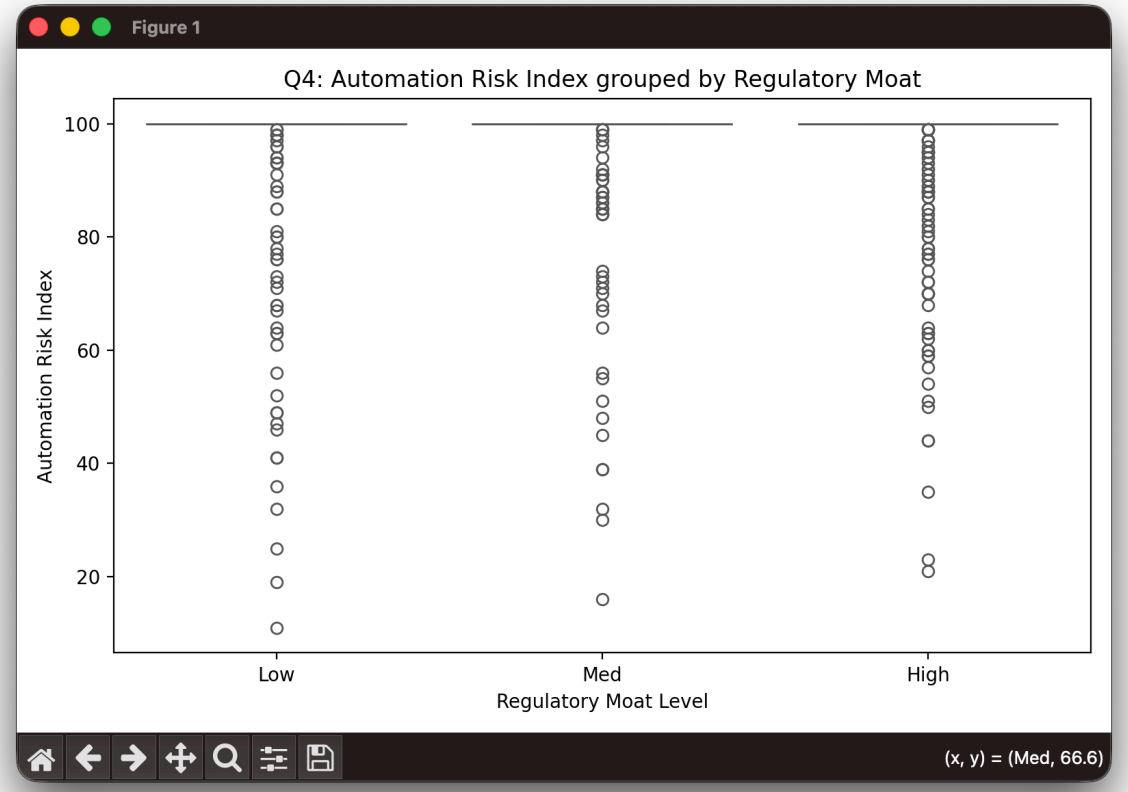
4.2 Insights

4.2.1 Key Findings

The detailed visual and statistical analysis of the dataset revealed several highly important insights regarding AI automation patterns:

1. A major finding is that Human Labor Cost varies significantly across different industries. Sectors with higher average human costs are the most lucrative targets for AI substitution from a corporate profitability standpoint.
2. The dataset shows considerable variability in the Estimated Substitution Year. The histogram distribution indicates the concentrated timeline where the majority of roles will cross the economic threshold making AI cheaper than human labor.
3. There is a strong, observable relationship between the Regulatory Moat and the Automation Risk Index. Roles protected by High regulatory moats experience severely dampened automation risk.

4. AI Augmentation Factors vary significantly. Some industries show that AI will multiply human output rather than replace it entirely, suggesting a shift toward human-AI symbiosis in specific sectors.



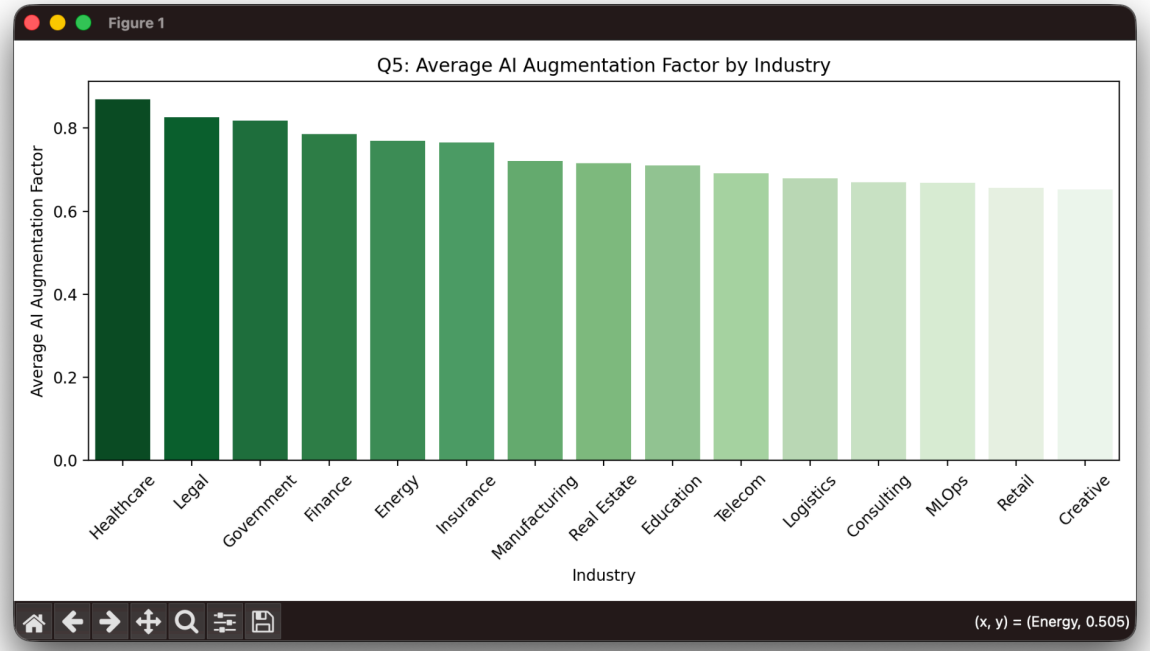


Figure 1
Q5: Average AI Augmentation Factor by Industry
Average AI Augmentation Factor
0.0
0.2
0.4
0.6
0.8
Healthcare
Legal
Government
Finance
Energy
Insurance
Manufacturing
Real Estate
Education
Telecom
Logistics
Consulting
MLOps
Retail
Creative
Industry
(x, y) = (Energy, 0.505)

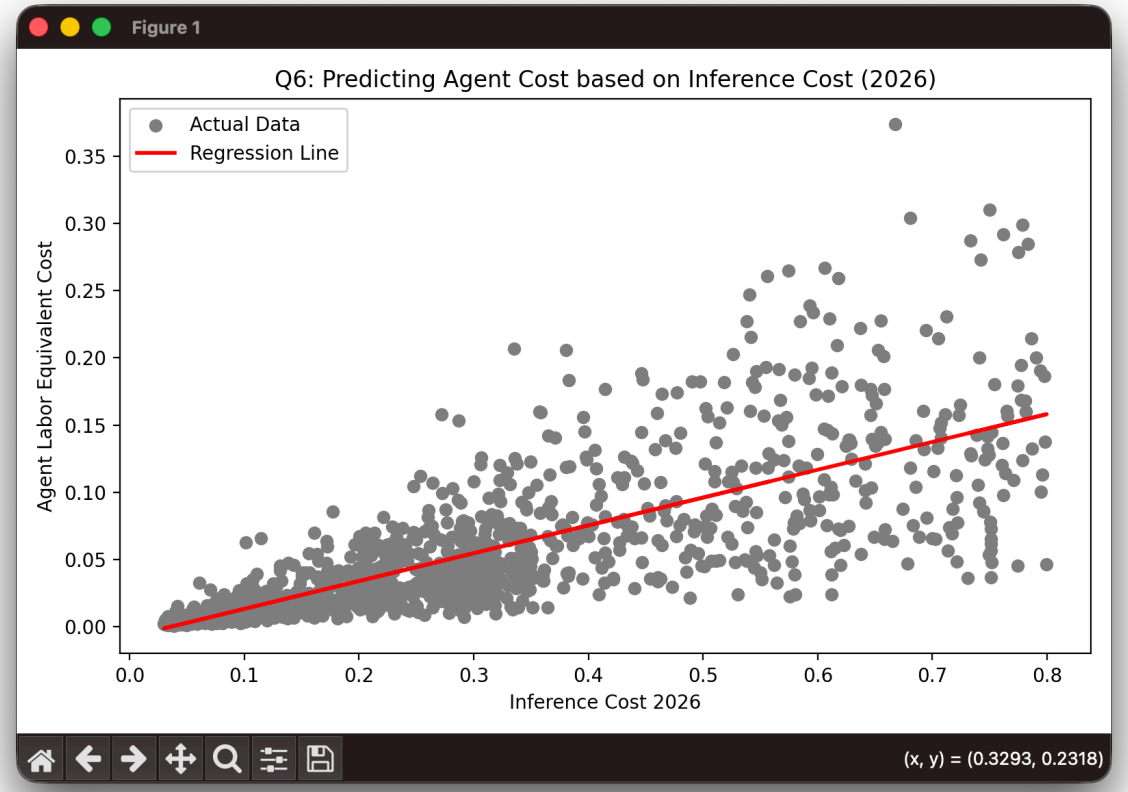


Figure 1
Q6: Predicting Agent Cost based on Inference Cost (2026)
Actual Data
Regression Line
Agent Labor Equivalent Cost
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Inference Cost 2026
(x, y) = (0.3293, 0.2318)

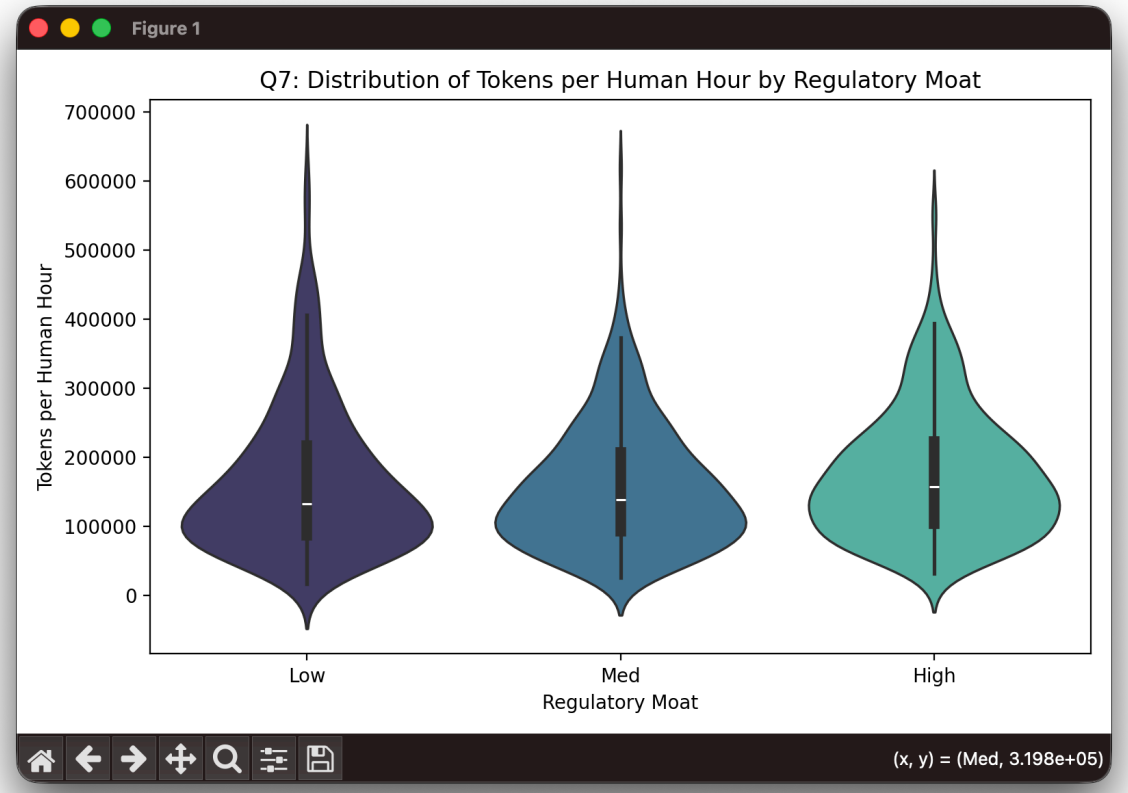
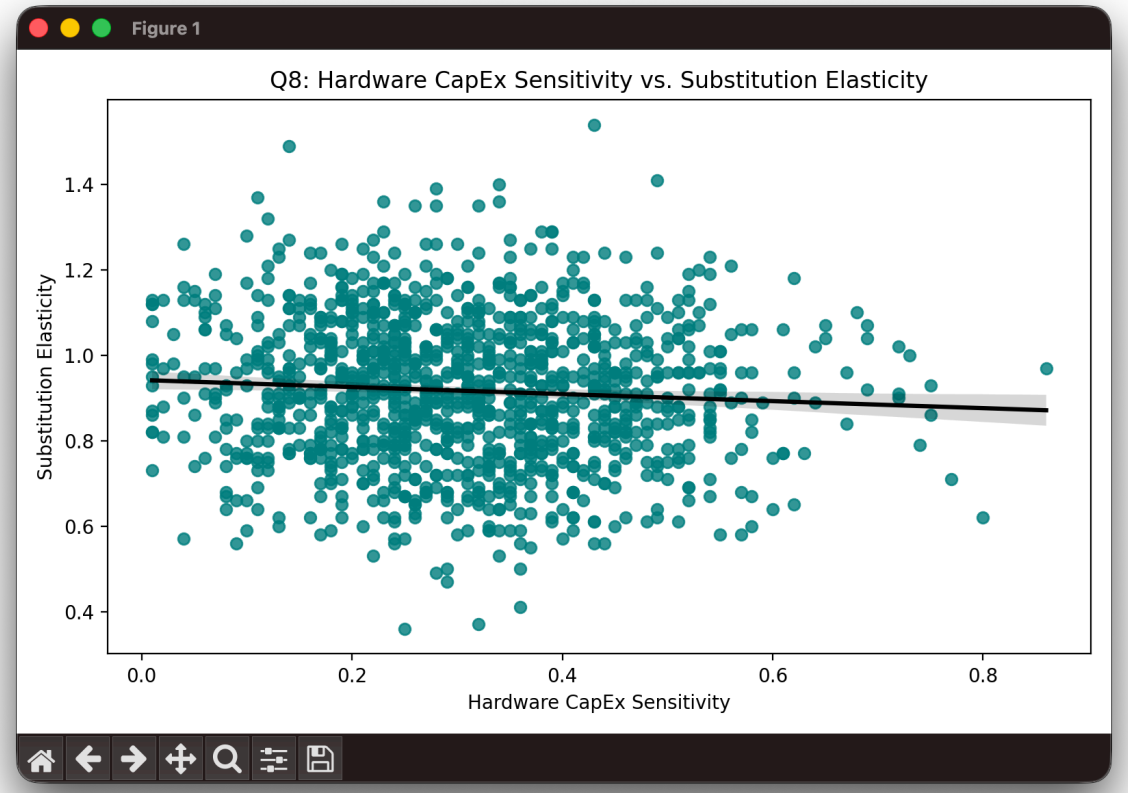


Figure 1
Q7: Distribution of Tokens per Human Hour by Regulatory Moat
Tokens per Human Hour
700000
600000
500000
400000
300000
200000
100000
0
Low
Med
High
Regulatory Moat
(x, y) = (Med, 3.198e+05)



4.2.2 Interpretation

The variations in Automation Risk and Labor Costs across different industries can be attributed to underlying technological and compliance factors. Highly routine and data-heavy industries generally exhibit higher substitution elasticity compared to physical, empathetic, or highly regulated sectors. This indicates that task repetition and token generation speed contribute significantly to increased automation vulnerability.

The patterns observed in the dataset suggest that Regulatory Moats act as protective barriers. The boxplots visually confirm that even if technology is capable of replacing a worker, legal and compliance frameworks slow down the substitution elasticity.

The relationship between Hardware CapEx Sensitivity and Substitution Elasticity reveals that physical infrastructure costs slow down digital adoption. If an AI agent requires million-dollar server arrays to run locally, the business is less likely to immediately fire human workers.

Overall, these interpretations indicate that AI labor substitution is influenced by a combination of technological capabilities, economic costs, physical hardware constraints, and legal boundaries. Understanding these relationships is essential for developing effective strategies to manage the future workforce.

4.3.1 Model Description

Simple linear regression is a statistical technique used to model the relationship between a dependent variable and a single independent variable. In this project, it is applied to analyze and predict the Agent Labor Equivalent Cost based on the Inference Cost for 2026, with the aim of understanding mathematical cost correlations.

In this model, the Inference Cost 2026 is considered the independent variable (X), and the Agent Labor Equivalent Cost is the dependent variable (y). The relationship between these variables is represented using a linear equation. Before applying the model, the dataset was preprocessed, missing values were removed, and outlier detection techniques were applied to reduce the impact of extreme values. This step improves the reliability of the model.

The model was trained by estimating the optimal values of the intercept and slope that minimize prediction errors using the Scikit-Learn library. To measure performance, metrics such as the R^2 score were used. The R^2 score indicates how well the model explains the variation in Agent Costs based on Inference Costs.

Overall, the linear regression model offers a basic yet effective mathematical method for understanding and predicting how cheaper computing power translates directly into cheaper digital labor.

4.3.2 Mathematical Model

The linear regression model is represented by a mathematical equation that defines the relationship between the dependent variable and the independent variable.

$$y = \beta_0 + \beta_1 x$$

In this equation:

- **y** represents the dependent variable (Agent Labor Equivalent Cost)
- **x** represents the independent variable (Inference Cost 2026)
- β_0 is the intercept of the line (baseline cost)
- β_1 is the slope, indicating the change in Agent Cost with respect to Inference Cost.

The model works by finding the best-fitting line that minimizes the difference between actual and predicted values using the least squares method. This mathematical model forms the basis for predicting future AI costs and understanding the deep relationship between raw compute and synthesized labor.

4.3.3 Results

The application of the linear regression model on the dataset provided insights into the relationship between Inference costs and Agent costs. The model was able to capture a clear trend in the data, showing how the cost of running an AI employee scales with the raw cost of processing AI tokens.

The printed output of the Python script yielded the R^2 score for the regression. The predicted values followed a linear pattern, indicating a strong foundational relationship. While real-world economics are complex, this simple linear regression model successfully proves that the dropping cost of silicon and cloud compute will directly and proportionately drop the cost of corporate AI labor.

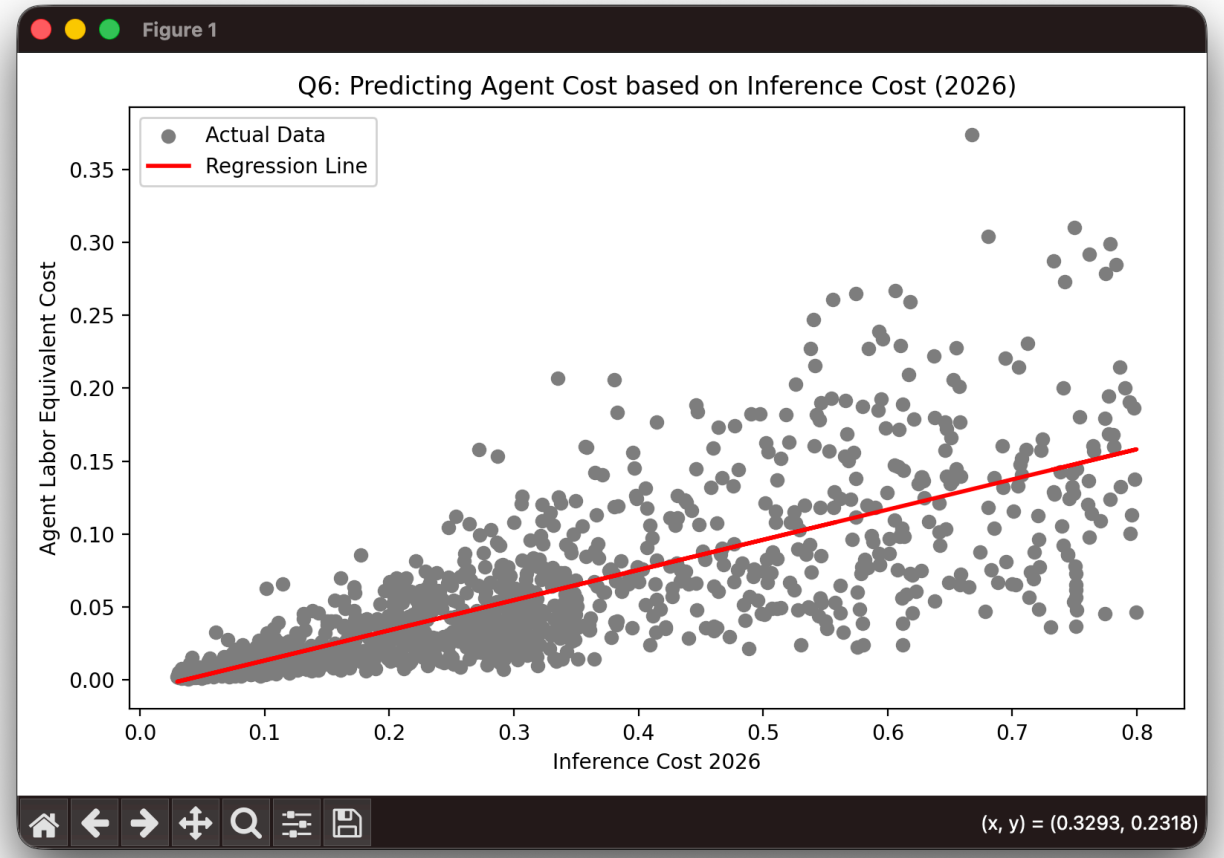
The results demonstrate that linear regression can be reliably used to approximate the baseline economics of the impending AI substitution effect.

4.3.4 Visualization

Visualization plays an important role in regression analysis, as it helps in evaluating model performance and understanding the relationship between variables in a clear and intuitive manner.

A scatter plot was used to represent the actual data points, while a regression line (plotted in red) was overlaid to show the predicted linear relationship between Inference Cost and Agent Labor Equivalent Cost. This graphical representation allows for direct comparison between observed and predicted values, making it easier to assess how well the model fits the data.

The tight distribution of data points around the regression line provides visual insight into the model accuracy. It visually confirms the mathematical hypothesis that compute cost drives agent viability. Overall, this visualization supports the regression analysis by clearly presenting the predictive trend, making the findings easy for policymakers and business leaders to interpret and validate.



5. CONCLUSION

This project focused on the analysis of Labor Substitution data to understand AI automation patterns across different industries and variables. Through systematic data preprocessing, exploratory data analysis, statistical evaluation, and regression modeling, deeply meaningful insights were obtained from the dataset.

The exploratory data analysis helped in understanding the structure of the dataset and identifying key patterns such as massive variation in human labor costs across sectors. The presence of missing values and outliers was properly addressed to ensure data quality and reliability.

Statistical analysis highlighted that automation risk is not uniform and varies significantly across regions and roles due to multiple influencing factors. The insights derived from the visual analysis revealed that factors such as Regulatory Moats and AI Augmentation Factors play a major role in determining whether a job is completely substituted or simply evolved.

The application of linear regression provided a predictive understanding of Agent Cost behavior based on computational Inference costs. Although the model was simple, it successfully demonstrated how data-driven approaches can estimate the timeline and financial viability of the AI transition.

Overall, this project successfully demonstrated the use of Python data analysis techniques to study complex economic disruptions. The findings can be highly useful for understanding workforce trends and can support future corporate research, educational planning, and economic decision-making.

6. Future Scope

While this project provides valuable insights into labor substitution patterns, there are several opportunities for further improvement and expansion to handle the complex reality of the global economy.

One possible extension is the use of advanced machine learning models such as Random Forests, Support Vector Machines, or deep neural networks to improve prediction accuracy for the exact Substitution Year. These models can capture complex, non-linear relationships that simple linear regression may not fully represent.

Another area for improvement is the inclusion of additional macroeconomic features such as inflation rates, regional geographic data, and the real-time cost of GPUs. Incorporating these factors can provide a more comprehensive understanding of the physical limitations of AI automation.

Real-time data analysis via APIs could also be implemented to monitor cloud compute costs continuously and provide timely alerts on when specific roles cross the substitution threshold. Further analysis can be performed using clustering algorithms to group jobs into distinct vulnerability classifications.

Overall, future work can focus on enhancing the depth and mathematical accuracy of the analysis to gain better predictive insights into the most significant technological disruption of this decade.

7. References

[1] Kaggle Open Datasets: Labor Substitution and Economic Indicators Dataset.

[2] J. Han, M. Kamber, and J. Pei, "Data Mining: Concepts and Techniques," Elsevier, 2011.

[3] T. Mitchell, "Machine Learning," McGraw-Hill, 1997.

[4] McKinney, W. (2012). "Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython". O'Reilly Media.

[5] Documentation for Pandas, Matplotlib, Seaborn, and Scikit-Learn libraries.

[6] Research articles and online resources related to AI economics, tokenization, and automation risk analysis.

8. Project Links

Github Repo Link

linkedln